Area Under Curve: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Area with the x-axis

Formulas

$$A = \int_a^b |f(x)|\,dx$$

KEY POINTS

- Area is always positive; $\int_a^b f\,dx$ is an algebraic (signed) area
- f changes sign at c, d: $A = \int_a^c f - \int_c^d f + \int_d^b f$
- Use symmetry (even function, periodic arches) to shorten the work

Watch out

Never integrate across a root of f without splitting

2. Area between two curves

Formulas

$$A = \int_a^b |f(x) - g(x)|\,dx$$

STEPS

1. Sketch the curves and shade the region
2. Solve the equations together for the intersection points
3. Integrate (upper − lower); split where the curves cross or a boundary changes

KEY POINTS

- Curves touching ($D = 0$) give a single meeting point, not a crossing

3. Strips: vertical or horizontal

Formulas

$$A = \int (y_{\text{upper}} - y_{\text{lower}})\,dx = \int (x_{\text{right}} - x_{\text{left}})\,dy$$

KEY POINTS

- Vertical strip: boundaries as $y = f(x)$; horizontal strip: boundaries as $x = f(y)$
- Choose the strip that needs fewer splits ($y^2 = 4ax$, $x = f(y)$, inverse functions: horizontal)
- Area between curve and y-axis from $y = c$ to $y = d$: $\int_c^d |x|\,dy$

4. Graphs you must sketch fast

KEY POINTS

- Polynomials: roots, multiplicity, end behaviour, turning points ($f' = 0$)
- Conics: $x^2 + y^2 = r^2$, $y^2 = 4ax$, $x^2 = 4ay$, $\frac{x^2}{a^2} \pm \frac{y^2}{b^2} = 1$; shifted forms after completing squares
- Trig: $\sin, \cos$ period 2π; $\tan, \cot$ period π; asymptotes of $\tan, \sec$ at $(2n+1)\frac{\pi}{2}$
- ITF: principal ranges $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $[0, \pi]$, $(-\frac{\pi}{2}, \frac{\pi}{2})$, $(0, \pi)$
- e^x and $\ln x$ are mirror images in $y = x$; $[x]$ steps; $\{x\}$ saw-tooth of period 1

5. Transformations and curve sketching

KEY POINTS

- $f(x \pm a)$: shift left/right; $f(x) \pm a$: up/down; $|f(x)|$: fold up; $f(|x|)$: mirror the right half
- $|x| + |y| = a$: square, area $2a^2$; $|x - h| + |y - k| = a$ same area
- Sketch with domain, symmetry, intercepts, sign of f', extrema, asymptotes, loops
- $y^2 = g(x)$: symmetric about the x-axis, exists where $g \geq 0$; area $= 2\int \sqrt{g}\,dx$

6. Standard results

Formulas

$$y^2 = 4ax,\ x^2 = 4by:\ A = \frac{16ab}{3} \qquad y^2 = 4ax,\ y = mx:\ A = \frac{8a^2}{3m^3}$$

$y = ax^2 + bx + c$ and a line meeting at α, β: $A = \dfrac{|a|(\beta - \alpha)^3}{6}$

KEY POINTS

- Ellipse πab; circle πr^2; parabolic segment $= \frac{2}{3}$ of its rectangle
- Two parabolas $y = a_1x^2 + \ldots$, $y = a_2x^2 + \ldots$: use $|a_1 - a_2|$ in the $\frac{(\beta-\alpha)^3}{6}$ result

7. Regions given by inequalities

STEPS

1. Replace each inequality by an equation and draw it
2. Test a point (e.g. origin) to choose the side
3. Take the common region, find corner points, integrate (split if needed)

Watch out

min{} / max{} or | | in a bound: draw both pieces and keep the right one

8. Shifting of origin

KEY POINTS

- Area is unchanged by $X = x - h$, $Y = y - k$
- Shift every curve, line and limit; the x-axis becomes $Y = -k$, the y-axis $X = -h$
- Then use standard results ($16ab/3$, $8a^2/3m^3$, πab)

9. Images and inverse functions

Formulas

$$\int_a^b f\,dx + \int_{f(a)}^{f(b)} f^{-1}\,dx = bf(b) - af(a)$$

KEY POINTS

- Reflection keeps area: $y = x$: (y, x); $y = -x$: $(-y, -x)$; $x = a$: $(2a - x, y)$; $y = b$: $(x, 2b - y)$
- Area under f^{-1} from $x = p$ to q = area between f and the y-axis from $y = p$ to q
- Area between f and $f^{-1} = 2\times$ area between f and $y = x$ (they meet on $y = x$ when f is increasing)

Watch out

Never try to write f^{-1} explicitly for $x^3 + 3x + 2$ type functions; integrate f instead

10. Variable area

Formulas

f monotonic: area of f, $x = a$, $x = b$, $y = f(c)$ least at $c = \dfrac{a + b}{2}$

KEY POINTS

- Area depending on a parameter k: write $A(k)$, then use $A'(k) = 0$ or a perfect square
- Area given, parameter unknown: write A in terms of the parameter and solve $A =$ given value; check the sketch for each root
- Area $A(x) = \int_0^x f$ given as a function: $f(x) = A'(x)$

11. Average and RMS value

Formulas

$$f_{\text{av}} = \frac{1}{b-a}\int_a^b f\,dx \qquad f_{\text{rms}} = \sqrt{\frac{1}{b-a}\int_a^b f^2\,dx}$$

KEY POINTS

- Average can be negative or zero; on $(0,\infty)$ use $\lim_{b\to\infty}\frac{1}{b}\int_0^b f$
- Average w.r.t. x^2: $\frac{1}{b^2-a^2}\int y\,d(x^2)$

12. Parametric curves

Formulas

$$A = \left|\int_{t_1}^{t_2} y(t)\,x'(t)\,dt\right|$$

KEY POINTS

- Convert x-limits to t-limits; closed curve: one full round
- $x = a\cos t$, $y = b\sin t$ gives πab

13. Common mistakes checklist

STEPS

1. Did you draw the figure and mark every intersection?
2. Is the integrand (upper − lower) positive on the whole interval?
3. Did you split at sign changes of f and where the boundary changes?
4. After shifting the origin, did you shift the axes and the limits too?
5. Region asked: “enclosed”, “above x-axis“, ”first quadrant“, ”smaller part“?
6. Answer in square units and positive